

These interview questions are based off of ones used by P. J. C. Reimer et al. in their paper:

P. J. C. Reimer, A. Gupta, C. Guastavino, and M. M. Wanderley, “User Experience with Digital Musical Instruments: A Transferable Method for Longitudinal Evaluation,” *J. New Music Res.*, vol. 53, no. 5, pp. 344–368, Oct. 2024, doi: 10.1080/09298215.2025.2540433.

Their data can be seen at <https://gitlab.com/creimer/longitudinal-evaluation-of-dmis> .

Entrance Interview

1. Why [don't you currently use DMIs/EMIs in active performance]? What would it take for you to use them in performance?

P1.

00:00:44 Speaker 2

Why not?

00:00:45 Speaker 2

Not too much really if it's playable enough, meaning that I, you know, I can get my heads around

00:00:58 Speaker 2

how to create sounds with it and to an extent where it becomes a tool for performance.

00:01:07 Speaker 2

So if I can acquire a certain, just a decent mastery or understanding of how it works and it does work, yeah, up to that extent, then that's all I need really.

00:01:28 Speaker 2

I mean, I'm thinking also that the other thing is that it just needs to fit in the rest of the gear that I have, for example, so, but that's more practical talk, I guess.

00:01:42 Speaker 2

if it's if it's a medium thing, for example, does it have just the, yeah, does it have the right MIDI interface or, yeah, cable, you know, there's MIDI A, MIDI B, yeah, there's, okay, just,

P2.

0:00:49 Speaker 2

I would say I would want someone to teach me and like to know how to use the

00:00:58 Speaker 2

the programs or use the synthesizers because I think synthesizers are fascinating.

00:01:04 Speaker 2

And also like sampling and how you can work with it.

00:01:08 Speaker 2

But I just, I don't know which tools to use and I don't know how.

00:01:13 Speaker 2

And also I'm a bit afraid of technology, but that's a personal issue.

P3.

00:00:33 Speaker 2

I mean, if I see some potential usage, maybe like different way of playing cello, for example, I would be interested, but not in generic way, like, I don't know, if it wouldn't be exactly, yeah.

a. Could you elaborate on the other contexts in which you use DMIs/EMIs?

P1.

00:02:13 Speaker 2

I would definitely, I mean, I use them definitely more at home than performances.

00:02:24 Speaker 2

but I guess the two are linked or ideally, but also maybe in some production.

00:02:33 Speaker 2

So like, music produced, music that's produced beforehand, not live in the performance.

P2. N/A

P3.

00:01:14 Speaker 2

Yeah, or like garageband or, like things like that.

00:01:32 Speaker 2

I don't know, like I tried like a couple of times, but not persistently.

2. What is the name of the DMI/EMI that you use(d)?

P1.

00:03:03 Speaker 2

Well, okay, the first one is MIDI keyboards and interfaces.

00:03:39 Speaker 2

Arturia, yeah, MIDI sequencer.

00:03:51 Speaker 2

And I've also been a big fan of the Novation Circuits.

00:07:10 Speaker 2

I like that it's a cello [referring to using electric cello].

P2.

00:04:33 Speaker 1

Maybe let me just go back to like, so did you have like a pedal?

00:04:39 Speaker 2

Yeah, just a foot pedal.

00:01:44 Speaker 2

His name is who Hugi Guðmundsson.

00:02:12 Speaker 2

But the piece's name was Veris.

P3.

00:01:03 Speaker 2

I mean, but I tried, but with that doesn't like the iPad things, I don't know if like apps for doing music counts or what.

00:01:14 Speaker 2

Yeah, or like garageband or, like things like that.

3. How long have you used it for?

P1.

00:06:30 Speaker 2

Well, that was, I don't know, about two years where I was doing that.

P2.

00:02:27 Speaker 2

I practiced with it for maybe three months, I guess.

00:02:34 Speaker 2

But not like every single day, but I had to rehearse using the pedals and yeah.

P3.

00:01:32 Speaker 2

I don't know, like I tried like a couple of times, but not persistently.

00:03:35 Speaker 2

Maybe over [the span of] a year.

00:02:30 Speaker 2

Perform with that, I feel like it's just fun, but yeah, I know it's different from practicing physical instruments for a lot of hours.

4. During a typical performance, what percentage of time do you use it?

P1.

00:06:50 Speaker 2

Most of it, 80%.

P2.

00:02:51 Speaker 2
I'd say maybe 10% or 15%.

P3. 0%

5. How many times a year do/have you used it in performance?

P1.
00:07:01 Speaker 2
Like 15.

P2.
00:03:14 Speaker 2
Over one year, or like.

P3. 0

6. What do you like about this instrument?

P1.
00:07:10 Speaker 2
I like that it's a cello.
00:07:30 Speaker 2
So what I like, essentially what I like about it is that I can play with pedals on it.
00:07:40 Speaker 2
So it doesn't have any bleeding, acoustic sound.
00:07:45 Speaker 2
So the sound that you're getting from it is actually something you built yourself electrically or electronically or digitally as opposed to the acoustic cello.
00:08:00 Speaker 2
I also like, in this one in particular, I like the fact that I can play standing up and so the ergonomically speaking, I think it's pretty cool.
00:08:09 Speaker 2
Which is, it's actually the reason why I got this one and nice.

P2.
00:03:29 Speaker 2
I liked that it was kind of not the same, or like it is the same instrument because I was still playing the cello, but it's a bit different.
00:03:42 Speaker 2
So you get a distance from what you know of it and like maybe the stigma of the classical aspect of it.
00:03:51 Speaker 2

but you are still using what and use your technology, use the technique you've learned, but you get to add another thing that takes off the pressure of something else.

00:04:01 Speaker 2

And you can add other stuff.

00:04:04 Speaker 2

And it's even, even though it's precise, and I had to be precise when I triggered the cues, but it's still the thing that the audience doesn't care if it's

00:04:18 Speaker 2

different or I can imagine that if I played it in one concert and it was a bit different or the electronics would have done something else it's okay or because yeah.

P3.

00:02:25 Speaker 2

I mean, it was just for fun.

7. What factors influence you to stop using a DMIs/EMIs?

P1.

00:08:29 Speaker 2

If it breaks, this guy [referring to electric instrument] just broke.

00:08:54 Speaker 2

And now, okay, like if, well, if I get something new, for example, then I might focus my attention onto that thing because, oh, it's actually, or, you know, because I don't have enough time to do

00:09:08 Speaker 2

that new thing and the other older things that I have.

00:09:49 Speaker 2

But it turns out that this new thing is just, yeah, it takes over because in the features that it has, it just sort of overlaps with the old thing, and then you just stop using the old thing.

P2.

00:05:10 Speaker 2

I think maybe it's a bit of a fuss.

00:05:24 Speaker 2

connecting the right stuff, making sure that everything is connected and working properly.

P3.

00:02:25 Speaker 2

I mean, it was just for fun.

00:02:28 Speaker 2

I mean, I didn't see like

00:02:30 Speaker 2

Perform with that, I feel like it's just fun, but yeah, I know it's different from practicing physical instruments for a lot of hours.

8. On average, how long do you typically use a DMI before retiring it? (Can be open ended)

P1.

00:10:39 Speaker 2

Yeah, no, actually, I remember I had a looper that I parted ways with.

00:10:45 Speaker 2

I used it for two years, maybe.

P2.

00:05:46 Speaker 2

Yeah, like, yeah, I was rehearsing the piece for a year, but I didn't add the electronics until...

00:05:53 Speaker 2

three or four months before the performance.

P3.

00:03:35 Speaker 2

Maybe over a year.

9. How often does your musical performance setup change?

a. Once a month

b. Once a year

P1.

00:11:27 Speaker 2

You know, I would say a couple years.

c. Once every 5 years

P2.

00:06:19 Speaker 1

So basically, once every five years.

00:06:23 Speaker 2

Yeah.

00:06:38 Speaker 1

But you say that you want to have a little bit more sometimes, those opportunities.

00:06:49 Speaker 2

Yeah.

00:06:46 Speaker 1

So you would want it to be more often.

00:06:49 Speaker 2

Yeah.

00:06:52 Speaker 2

I would love to be able to also because I sing and I think it would be awesome to be able to do some manipulate or do some stuff and then manipulate it in the computer or have some synthesizers to do something different, add layers to it.

P3.

00:04:03 Speaker 1

It's mostly just cello.

10. Are you familiar with the theremin, laser harp, or similar instruments to those? (Can be open ended)

P1.

00:11:54 Speaker 2

Theremin, yeah.

P2.

00:07:25 Speaker 1

Theremin, but laser harp, not so much.

P3. Theremin

a. Have you performed with them previously?

P1. No

P2. No

P3. No

b. Have you watched someone else perform with them?

P1.

00:12:06 Speaker 2

Not live, no.

P2.

00:07:38 Speaker 2

Not live, but I've looked at videos and now.

P3.

00:04:42 Speaker 2

No, I just watched some good videos on YouTube.

c. Do you know how to use them?

P1.

00:12:16 Speaker 2

Not really.

P2.

00:07:50 Speaker 2

I've never had contact with them myself, but I kind of know that one hand has to, is doing one thing and the other is controlling the pitch.

P3.

00:05:02 Speaker 2

I mean, there's like the movement of the hands that there's some magnetism thing I guess.

d. Are you familiar with any previous research conducted related to those instruments?

P1. No

P2. No

P3. No.

11. If you have used these instruments (theremin, laser harp, etc.) before, did you create your own mapping for the instrument? (Can be open ended)

a. If so, please describe how you approached the process of creating a mapping.

b. How challenging did you find the process of creating your mapping?

c. How-time-consuming did you find the process of creating your mapping?

d. How did the process of developing your own mapping affect your approach to

playing the theremin or laser harp?

Free exploration / Tasks

1. Play with just the IMU bow.
2. Play on all of the strings.
3. Play with the fingers past the fingerboard.
4. Change a scala scale and play.

P1. Angle to play on A string is tough.

P2. "Nice not having to press anything [with the right hand]".

Evaluation Session Interview

1. Please provide a quick summary of what you did with the instrument during the practice today.

P1.

00:00:37 Speaker 2

There was, there were a few different settings that we tried, I think, two or three.

00:00:46 Speaker 2

... the first experiment was with the bow [IMU], so I tried the full thing basically, right, the final prototype, which was pretty cool.

P2.

00:00:16 Speaker 2

Yeah, just you showed me a bit of round and how it functions and then I just got to sit down and explore a bit how it works and try it.

00:00:30 Speaker 2

Try it out and find out and get to know it myself.

P3.

00:00:29 Speaker 2

Can experiment like different movements, how they respond to the instrument.

00:00:36 Speaker 2

And try to understand what if I press something, what happens.

00:00:43 Speaker 2

And I don't know, yeah, trying different techniques, ways, yeah.

a. How much time did you spend practicing with the instrument?

P1.

00:01:13 Speaker 2

30 minutes.

P2.

00:00:51 Speaker 2

Well, it just slipped away.

00:01:02 Speaker 2

15 or 20.

P3.

00:00:56 Speaker 2

I don't know how much time I did.

00:01:03 Speaker 1
Or 20, no, about 20 minutes.

2. Please elaborate on your response to the statement “I felt in control of the instrument.”
a. What parameters are you trying to control? Any specific technique?

P1.

00:03:46 Speaker 2
I feel as a result of this whole experiment that I wasn't really in control of the pitch.
00:03:51 Speaker 2
But maybe that's okay because I was in control of other things.
00:03:56 Speaker 2
Like the bow thing is pretty cool.
00:04:01 Speaker 2
Also to an, like not, it doesn't mean that I knew if I wanted to play the A string, oh, I, you know, I could, I knew where to put my hand and it would play the A string.
00:04:10 Speaker 2
But at least the movement of my hand,
00:04:14 Speaker 2
led to some change in the sound, and that's what matters, maybe.
00:04:18 Speaker 2
So it's the same with the pitch of the notes, actually.
00:04:22 Speaker 2
The movement of my hand on the string was creating something.

P2.

00:01:30 Speaker 2
I mean, it was not as similar as the control that you have with the bow because you would have no reference.
00:01:36 Speaker 2
But I still felt like I had some, like I had control over the pitch.
00:01:42 Speaker 2
And once you got to know like with the distance and that you had to move your whole hand, then I felt more in control.
00:01:49 Speaker 2
Like then I was in charge kind of.
00:01:56 Speaker 2
And when I maybe just tried to use the bow hand a bit, and the concept is in itself easy.
00:02:12 Speaker 2
It's like the A string and the C string are definitely the easiest because they are, you can go and they exaggerate, but I was really, I really wanted to try to get a hold of the D and the G.

00:02:27 Speaker 2

And I think I managed for a bit there.

00:02:30 Speaker 2

But that's of course more difficult because then you have to be more precise.

00:02:56 Speaker 2

or what I try to do is try to find other, other anchors, which other anchors can you find?

00:03:05 Speaker 2

And then you try to use the hand to go higher or lower and try to find which, I don't know, I think just finding another anchor and feeling it in your body.

00:03:17 Speaker 2

And then, I mean, if you could practice it more than I think you could.

P3.

00:01:24 Speaker 2

I would disagree a bit because I didn't know if I was doing something.

00:01:32 Speaker 2

if I was really controlling something or it was just random things.

a. What effects are you trying to achieve?

P1.

00:04:47 Speaker 2

I don't think I was trying to do anything in particular.

00:04:55 Speaker 2

I was trying to get a feel for what it can do.

00:05:04 Speaker 2

I could say that I was trying to play a scale in the beginning, but I quickly dropped that.

P2.

00:03:34 Speaker 2

I think that would have like come a bit later in the process when I was like in like getting the hang of it.

00:03:44 Speaker 2

That's why I wanted to try like the vocal fry and then using the and then using like voice or like and it was way different.

00:03:54 Speaker 2

And I can see a potential like for movie sounds, but at first I was trying to do maybe a trill or something, but it made a trill almost by itself, or because it's a bit like... [sensors are not so stable]

P3.

00:01:43 Speaker 2

Yeah, kind of like the tune, like having a reference, some tones that I would like to hear, because when you press the finger, you associate with the tone, kind of.

00:02:04 Speaker 2

I don't know, some rhythms combined with some tones, but yeah, it was hard to achieve this.

c. Please elaborate about the mapping, interface, and sound.

i. Do you feel that you are able to produce sounds that you find pleasing or useful in a musical context with this instrument?

P1.

00:05:22 Speaker 2

Yeah, I do.

00:05:26 Speaker 2

Useful at least, yeah.

P2.

00:04:33 Speaker 2

Yeah, definitely.

P3.

00:02:33 Speaker 2

I don't know, because since I cannot control very well, I wouldn't know how to use them because I wouldn't feel comfortable using a concert because, yeah, I don't know how to use them.

ii. Do you think the mapping is easy to learn?

P1.

00:05:36 Speaker 2

Well, insofar as it's learnable.

P2.

00:04:46 Speaker 1

Okay, you kind of answered it.

P3.

00:02:57 Speaker 2

I don't think so.

00:03:00 Speaker 2

Because, yeah, I still don't know how it actually works.

iii. Is the mapping intuitive?

P1.

00:05:50 Speaker 2

Yeah, also, I would say, yeah, it doesn't mean that it's working, but it should be, exactly.

P2.

00:05:24 Speaker 2

I guess if you rehearse it, or like if you rehearse and get to know it a bit better, like

00:05:32 Speaker 2

You can't tell, I think, in 15 minutes, but I definitely think you, in those 15 minutes, I got the hang of some of the things, which tells me that with a bit of practice and a bit more mapping, then you would be able to both have more control and play it more.

P3.

00:03:10 Speaker 2

No, because with the it was really hard to understand how it works.

00:03:15 Speaker 2

I mean, somehow, at the end I managed, but if you wouldn't say you have to do this and this, I wouldn't figure it out.

iv. Is the interface easy to control?

P1. No

P2.

00:06:00 Speaker 2

I wouldn't say it's easy to control.

P3.

00:03:44 Speaker 2

But what's the difference between interface and mapping?

00:03:51 Speaker 2

Okay, so it's the same answer. [No]

3. Please describe any other notable aspects (e.g. breakthroughs, frustrations, etc.) of your experience using the Augmented cello over this time period.

P1.

00:06:18 Speaker 2

Breakthrough.

00:06:28 Speaker 2

No, I think I could totally see that in a, yeah, in a pedal board environment with like more effects.

00:07:02 Speaker 2

So I think for this sound,

00:07:05 Speaker 2

It would be really cool.

P2.

00:06:23 Speaker 2

I think maybe the first, like the first minutes are always a bit rough, right?

00:06:34 Speaker 2

Maybe the most frustrating part was trying to get to know the bow hand because that's the thing that's way more different.

00:06:44 Speaker 2

Or like you have to carry it way differently than you carry the left hand.

P3.

00:04:34 Speaker 2

I was trying to make some rhythms, but it didn't work so much.

00:04:43 Speaker 2

I mean, somehow, but you get frustration as well, so a little bit, yeah.

00:04:49 Speaker 2

I think the only thing I could control is like maybe stopping all the sound and doing the sound something.

Exit Interview (after the whole session, and exit questionnaire)

1. Please elaborate on your response to the statement “*The instrument allows me to do things I cannot do with acoustic instruments/other DMIs I have used in the past.*”

P1.

00:00:26 Speaker 1

I don't think I've ever heard this kind of sound than any other DNI or instrument.

00:00:30 Speaker 1

So, for one thing, just unique.

P2.

00:00:00 Speaker 1

... I just think this is um of course it's it's a different sound I mean uh and when you lose or like when you put down the bow you know it's it's just a completely

00:00:27 Speaker 1

the possibilities are different of course you don't have of course it's not the cello either like but it's its own thing um different possibilities allows for different interpretation um and I think if you would get like more control of it and like understand and get more hang of the uh the glove bow um

00:00:57 Speaker 1

then it's of course you can do stuff with it that you cannot do with acoustic instruments because it's also based on electronics and based on sensors it's not based on the bow being on the string.

P3.

00:00:27 Speaker 2

I think it's different from playing a normal instrument for me.

00:00:35 Speaker 2

You can move...

00:00:37 Speaker 2

make out more movements that you cannot do with just the normal instruments.

00:00:43 Speaker 2

And they make different responses.

2. Please elaborate on your response to the statement: *"I feel this instrument could be useful in my everyday musical practice."*

a. In what context will you be using the instrument?

P1.

00:00:51 Speaker 1

I wouldn't, okay, the thing is, I could use, I could see it as useful in a musical setup, but every day, it wouldn't help me in my everyday musical practice.

00:01:04 Speaker 1

It would be like a one-of-a-kind thing that you pull out of the drawer or something.

P2.

00:01:52 Speaker 1

I could see

00:01:53 Speaker 1

very beneficial use for it in like improvisation and using using it in like to get to know yourself as a musician better as well or like follow your impulses kind of and also partake in

00:02:19 Speaker 1

and maybe other projects that you wouldn't maybe do inside that conservatory or like you could maybe be more connected to like rhythmical or like experimental music practices which would widen your horizons which is why I think it would be beneficial.

P3.

00:01:06 Speaker 2

No, it wouldn't be useful because it's really hard to manage it, so I don't know how I would use it.

a. Along with which other instruments?

P1.

00:01:28 Speaker 1
Definitely drum kit or like pedals.

P2.

00:02:50 Speaker 1
I could see this pair like with if you have a just like electronic music or other instruments that have similar, not maybe similar, but do some electronic improvisation or yeah.

00:03:13 Speaker 1

I think it would be cool also with other student instruments, like if you could add the violin or something to have different, because the cello is more deeper, or like I guess it's different.

P3.

00:01:46 Speaker 2

I mean, I can't imagine if it's something more digitally that you can combine, but with another acoustic instrument.

00:01:52 Speaker 2

I don't know.

00:01:54 Speaker 2

It's a bit harder.

3. Is it novel?

P1.

00:01:44 Speaker 2

Yeah, I think so.

P2.

00:03:44 Speaker 2

Yeah, I would say so.

00:03:48 Speaker 2

I have at least never tried it before and I've never seen anything.

P3.

00:02:07 Speaker 2

I mean, yeah, I think so.

00:02:11 Speaker 2

I've never seen something like this, so.

4. Is there a sense of freedom? Does time seem to “stop”? (After you push the bluetooth pedal to record a sample)

P1.

00:02:37 Speaker 2

Yeah, in a way, because it's this granular, it's this grain looping over itself.

00:02:45 Speaker 2

I think, the nature of granular synthesis is that, I guess.

P2.

00:04:57 Speaker 2

Yeah, I would say so.

00:05:13 Speaker 2

But the feeling for me is when I think time stops, is when it's actually slipping away.

00:05:20 Speaker 2

Or it's like, because it all goes by so fast because you're so in the moment.

00:05:26 Speaker 2

And I think these minutes just completely slipped by, which means that for me, it was very much in the moment.

P3.

00:02:57 Speaker 2

Yeah, like not having to use the bow is interesting, like because you can experiment other movements of the hands.

5. On a scale of 1 (low) to 5 (high), how responsive do you find the system to be?

P1.

00:03:57 Speaker 2

I mean, you know, I'd make it 3 because the bow was like, yeah, there was something there and, yeah.

P2.

00:06:04 Speaker 2

3.5 to 4.

P3.

00:03:20 Speaker 2

Maybe 2?

00:03:24 Speaker 2

Yeah.

6. What is your reaction to changing the scala scales?

P1.

00:04:29 Speaker 2

I would never change scale while I'm playing.

00:04:45 Speaker 2

Or, yeah, maybe there's a case where it's like, actually, it could be really useful to do that, but I can't think of it.

P2.

00:06:24 Speaker 2

Sometimes I felt more of a difference, like I think I kind of felt more of a difference between the major and the minor than I did between, for example, the chromatic scale and the major one.

00:06:44 Speaker 2

But maybe it had something to do with the little sensory sounds or something.

P3.

00:03:39 Speaker 2

A bit interesting, but still confusing a bit.

7. Would you use any other scales/create your own?

P1.

00:04:59 Speaker 2

Yeah, I think, but that's obviously...

P2.

00:07:04 Speaker 2

I think it would be cool, but I really had to be in the mood for it.

00:07:33 Speaker 2

I think it would be cool because it gives you much artistic freedom.

00:07:37 Speaker 2

Both like both the element of you are sampling the sounds that you want, but then you can also create your own scale to add on it.

00:07:47 Speaker 2

So like if you do it correctly, you are using the creative energy in a very, very wide range, I would say.

P3.

00:03:53 Speaker 2

I don't know.

00:03:58 Speaker 2

I mean, I feel like the ones that already existed for me [were good enough]...

8. Does this instrument remind you of any others?

P1.

00:06:16 Speaker 2

I think it's trying to remind me of the Theremin.

P2.

00:08:11 Speaker 2

Maybe the theremin that we talked about.

00:08:16 Speaker 2

But it's only because of the bow or like the distance thing with the bow.

P3.

00:04:07 Speaker 2

I mean, since you cited the theremin probably somehow remind and yeah, don't have other preferences.

9. Would you plan on using this instrument in the future? In what context?

P1. "In a live performance, I would use this."

00:06:31 Speaker 2

If it gets, if it gets better.

00:06:51 Speaker 2

Or like I said, to create some, interesting soundscapes.

00:07:26 Speaker 2

Also, if you could access the parameters of the granular synthesis, I think that would be really cool.

P2.

00:08:38 Speaker 2

At this point in my life, I think no.

00:08:39 Speaker 2

But if I were playing the cello more and maybe more brave in the electronics world, then sure, yeah.

P3.

00:04:25 Speaker 2

I mean, if you work on that and you improve it, yeah, why not?

10. How long do you estimate you will be using the instrument?

P1.

00:09:09 Speaker 2

It would definitely be part of, if I, if I were to acquire it, would become part of my setup.

P2. N/A

P3.

00:04:51 Speaker 2

If improved, I think I would use it more, yeah.

00:05:24 Speaker 2

I feel like, yeah, maybe I could be a phase if it's, yeah, it depends, I guess.

11. Focus on one particularly negative experience you had during this process. Please describe your experience.

P1.

00:09:36 Speaker 2

Yeah, this normalization, pitch normalization, where it's like the fact that we didn't know what the pitch was or what was the root pitch of the sample and it wasn't mapping correctly to the fingers.

P2.

00:09:26 Speaker 2

I think in the beginning, I just have a really hard time not understanding.

00:09:39 Speaker 2

But it's like I've always had such a hard time not being good at something right from the start.

00:09:47 Speaker 2

Which is kind of ironic when I say that I play the cello or like, and I sing, which takes a lot of time.

00:09:54 Speaker 2

But it's also like, I know that I can be a bit frustrated.

00:10:02 Speaker 2

at first, when I'm like, and what am I supposed to do?

P3.

00:05:56 Speaker 2

Of this, yeah, the fact that I couldn't like understand what I was doing, like, yeah, I guess was doing random stuff.

12. Focus on one particularly positive experience you had during this process. Please describe your experience.

P1.

00:10:07 Speaker 2

the first time there were some cool things that we were able to achieve some, really nice stuff with the, without the bow and just sliding.

00:11:07 Speaker 2

And today the bow generally was pretty cool, I think.

P2.

00:10:23 Speaker 2

I felt it was really cool when I, like the vocal fry thing on the C string was really cool.

00:10:32 Speaker 2

And then when I like more sang and then recorded it and then it became so much more clearer.

00:10:41 Speaker 2

I felt that was like a key moment in my control of like how it sounded.

P3.

00:06:34 Speaker 2

Yeah, like not using the bow was interesting.

13. How did your engagement with the instrument develop over time?

P1.

00:11:47 Speaker 2

It's hard to tell.

00:11:48 Speaker 2

I was going to say, I feel like I, in the beginning, I was just like, holy **** what's happening?

00:11:55 Speaker 2

And then I got sort of better control and

00:12:01 Speaker 2

And then I understood what could come out of it.

00:12:03 Speaker 2

So you sort of change your perspective and you change your expectations.

00:12:08 Speaker 2

In the beginning, I wanted to play a scale, for example.

00:12:11 Speaker 2

And that's okay.

00:12:12 Speaker 2

It's really tough.

00:12:14 Speaker 2

Changing strings also was, you know, hard.

00:12:17 Speaker 2

So then that changes, that evolves.

00:12:24 Speaker 2

But then, no, the thing is, today we did something different.

00:12:27 Speaker 2

So it's also, it also generated some other

00:12:33 Speaker 2

Surprises and things like that.

P2.

00:11:13 Speaker 1

Smoother and smoother, I think, or like more understanding, I think.

P3.

00:06:47 Speaker 2

Yeah, maybe I start understanding how to move the hand without the bow a little bit more.

00:06:56 Speaker 2

with the left [hand], I think, I don't know, I didn't manage.

14. Did you feel as though your interactions with the instrument became more intuitive or automatic over the time spent with the augmented cello?

P1.

00:12:48 Speaker 2

Yes, to the extent, you know, that we just take one experiment at a time.

P2.

00:11:46 Speaker 2

Yeah, I would say so.

P3.

00:07:13 Speaker 2

Not that much.

13. Do you think this process (and the time you spent with the "Ghost" cello) was:

a. Rewarding?

P1.

00:13:10 Speaker 1

Yeah.

P2. Yes – Implied from other responses below.

P3.

00:07:27 Speaker 2

It's inspiring, I would say, like interesting, like, yeah.

b. Worthwhile?

P1.

00:13:15 Speaker 2

Yeah.

P2.

00:12:02 Speaker 2

I think it was totally worthwhile.

00:12:05 Speaker 2

And fun.

P3.

00:07:39 Speaker 2

Yeah, okay.

a. Frustrating?

P1.

00:13:27 Speaker 2

A little bit, but not much.

P2.

00:12:13 Speaker 2

Maybe a bit, but the enjoyment of it made-up for it, all can exist.

P3.

00:07:45 Speaker 2

I don't know, not part, I mean, I wasn't angry, like I didn't have any expectation, so I was okay.

14. Do you have any suggestions for improvement?

a. If you could change the sound, what changes would you make?

P1. Implied from the above comments.

P2.

00:13:36 Speaker 2

I don't think so.

P3.

00:08:12 Speaker 1

Better bow control.

00:08:15 Speaker 2

And tune control.

00:08:17 Speaker 2

Yeah.

00:08:26 Speaker 2

Yeah, volume, I wouldn't complain about that.

a. If you could change the mapping, what changes would you make?

P1.

00:13:51 Speaker 2

Like I said, I think it would be really cool to have a continuous scale, but I'm aware that it's probably...

P2.

00:12:41 Speaker 2

But the one thing that I felt was maybe just the sensors, or it would be very cool if the sensors were more accurate, because then I can see.

00:12:59 Speaker 2

Because you can feel that you're doing something, but you have to...

00:13:02 Speaker 2

do more drastic changes to the placement of your hand.

P3. Get rid of the top black backboard a the top (annoyed her head).

00:09:10 Speaker 1

so get rid of this top.

00:09:11 Speaker 2

But also what I, as I said before, like making, also the acoustic cage, you don't need it because it's...

00:09:22 Speaker 2

Like, yeah, it's just the string basically and...

11. Is there anything that you would like to add?

P1.

00:15:23 Speaker 2

No, I already told you.

P2.

00:14:02 Speaker 2

I thought it was really cool to not have a bow.

00:14:05 Speaker 2

Like when I put the bow down, I felt a bit more free.

00:14:11 Speaker 2

And because I mentioned with the dance thing, because I did ballet for many, many years.

00:14:20 Speaker 2

So I think it's also very cool that it connects or I can see the potential of it connecting

00:14:27 Speaker 2

more art forms together because like when you're free of the bow, you're freer of the restrictions.

00:14:40 Speaker 2

Yeah, I can just see the dance of it, which I think is cool.

P3.

00:10:08 Speaker 2

Yeah, I think you should keep going and yeah, it's interesting.